

INST 327: Database Design and Modeling

Final Project Report

December 13, 2023

Introduction

In response to the escalating significance of electric vehicles (EVs) and the evolving landscape of sustainable transportation, our team embarked on a comprehensive endeavor — the creation of the Electric Vehicle Population Database. With a focus on the state of Washington, this initiative aims to collate and organize pertinent data on Battery Electric Vehicles (BEVs) and Plug-in Hybrid Electric Vehicles (PHEVs) registered through the Washington State Department of Licensing (DOL). This database serves as a dynamic repository, capturing essential details such as vehicle make and model, registration locations, electric vehicle types, and eligibility for the state's Clean Alternative Fuel Vehicle program.

As the automotive industry undergoes a transformative shift towards cleaner and greener technologies, understanding the nuances of EV adoption, market trends, and the geographical distribution of these vehicles becomes imperative. The Electric Vehicle Population Database is positioned to offer a nuanced exploration of the electric vehicle landscape in Washington. Beyond the fundamental attributes, the database incorporates additional data points like Vehicle Identification Numbers (VIN), model years, electric range, and DOL license numbers. By addressing the intricate relationship between VINs and locations, it aims to provide a holistic perspective on the dynamic EV market.

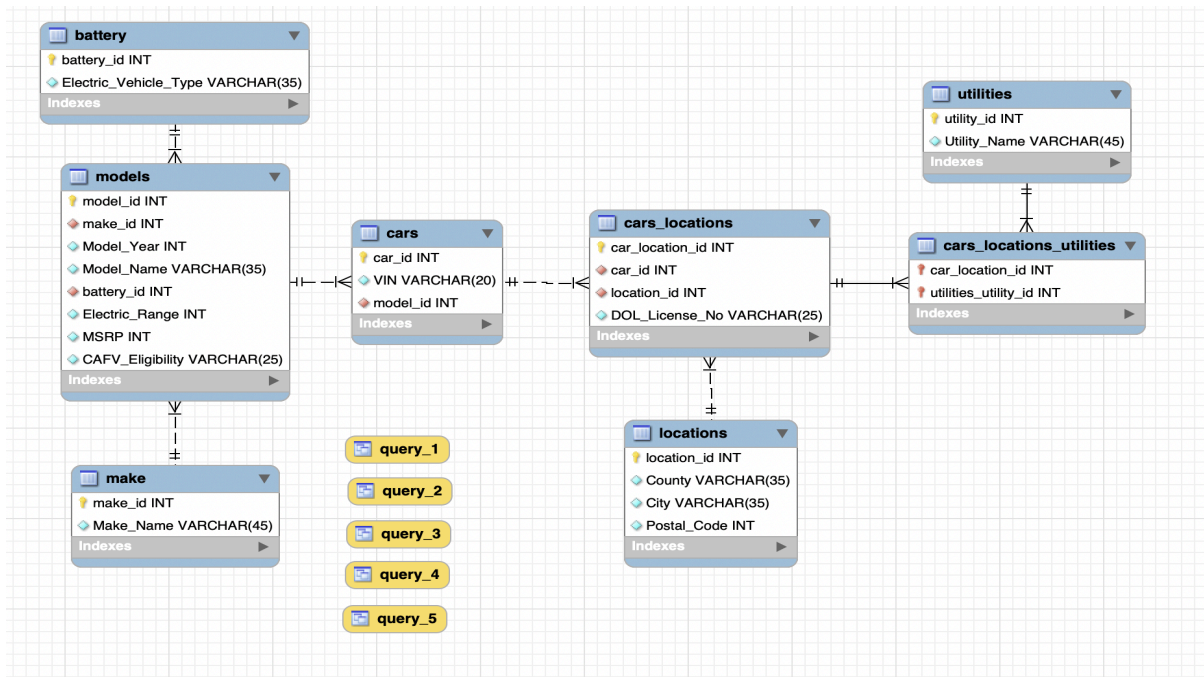
This initiative not only caters to the curiosity of researchers and policymakers but also holds substantial value for local businesses, electric vehicle charging providers, and environmental advocacy groups. By offering a detailed insight into the historical and current landscape of electric vehicles in Washington, the database strives to foster informed decision-making and facilitate a sustainable future for transportation.

Moreover, recognizing the sensitive nature of personal information related to vehicle owners, our commitment extends to ensuring robust data security, adherence to privacy regulations, and ethical handling practices. Regular updates and collaboration with relevant authorities are integral to maintaining the accuracy and relevance of the database, ensuring its enduring utility for stakeholders invested in the future of electric mobility..

Database description

Our database is built upon a dataset provided by the Washington State Department of Licensing, focusing on electric vehicles registered in the state of Washington. The original dataset consisted of over 120,000 entries, and contained various details about each car such as VIN, location, MSRP, electric range, legislative district, and utility providers. However, to streamline functionality and eliminate unnecessary complexity, we made strategic reductions. The redefined database now includes essential information like VIN, location, utility companies, make and model details, electric vehicle type, electric range, MSRP, and eligibility for the state tax credit (CAFV). Additionally, we've implemented a privacy-enhancing measure by assigning each car a unique 'car_id', affording the opportunity to censor sensitive information such as VIN numbers. This approach provides an extra layer of confidentiality to the database, ensuring the protection of private details while maintaining the integrity and usability of the dataset. This approach enhances database efficiency by eliminating redundant and unnecessary data. Notably, our database is designed to accommodate many-to-many relationships, allowing a car to have multiple locations, and consequently, multiple license/registration numbers, creating a seamless link with various utility providers. While the current data in our database is limited for project purposes, the structure is scalable, demonstrating its potential to become a more organized and usable alternative to the original dataset with a more extensive collection of entries.

Updated ERD



Sample Data

Our database contains these 8 tables:

1. make

- Contains information about the different car Makes that are registered in WA
- This table has only 10 rows, since only 10 makes were present in the dataset at the time of the creation of this database

Make:

make_id	Make_Name
1	BMW
2	CADDILAC

2. model

- Contains information about the different Models of cars that are registered in WA
- make_id and battery_id are foreign key references in this table

Model:

model_id	make_id	model_year	model_name	battery_id	electric_range	MSRP	CAFV_eligibility
1	1	2016	330E	2	14	43700	Not eligible
13	5	2016	SOUL	1	93	31950	Eligible

3. battery

- a. Contains information about the different electric vehicle types
- b. This table currently has 2 options: Battery Electric Vehicle (BEV) or Plug-in Hybrid Electric Vehicle (PHEV)
 - i. More can be added if / when more electric vehicles come to market

Battery:

battery_id	electric_vehicle_type
1	Battery Electric Vehicle
2	Plug-in Hybrid Electric Vehicle

4. cars

- a. Contains information about specific cars that are registered in WA
- b. model_id is a foreign key reference in this table

Cars:

car_id	VIN	model_id
1	2C4RC1H70K	11
21	5YJRE11B08	25

5. locations

- a. Contains information about different locations throughout WA

Locations:

location_id	County	City	Postal_Code
1	Asotin	Asotin	99402
110	King	Seattle	98101

6. cars_locations

- a. Helps identify the many-to-many relationship between a car and locations
- b. car_location_id is the primary key, and car_id and location_id are foreign key references
- c. It also contains DOL License Numbers for each car at that location

Cars locations:

car_location_id	car_id	location_id	DOL_license_no
3	3	284	211241698
4	3	185	255619941

7. utilities

- a. Contains information about utility companies throughout the state of WA

Utilities:

utility_id	utility_name
1	AVISTA CORP
3	INLAND POWER & LIGHT COMPANY

8. cars_locations_utilities

- a. Helps identify the many-to-many relationship between a car/location and the utility company it uses
- b. car_location_id and utility_id are composite primary keys in this table

Cars locations utilities:

car_location_id	utility_id
2	2
2	8

Queries

** Please see inst327 final queries.sql file to view the queries **

	Req. A	Req. B	Req. C	Req. D	Req. E
query_1	X	X			
query_2	X	X	X		
query_3	X	X	X		
query_4		X	X		X
query_5	X		X	X	

Questions the queries aim to answer

1. What models of electric vehicles are currently registered in Washington?
2. What electric vehicle models have more than one registration in WA, and what is the number of registrations it has?
3. How many electric vehicles registered in WA are battery powered?
4. What is the average number of locations for each EV in WA?
5. What is the number of cars per city, and what is the average electric range of those vehicles?

Changes from original design

Our initial proposal, as well as the next few submissions, were based on the idea that we would only be selecting unique Vehicle Identification Numbers (VIN) from the dataset, meaning each VIN would only appear once in the entirety of our database. After speaking with the teaching assistants, we as a group came to the conclusion that it would be a better choice to leave in duplicate values for VIN numbers. These duplicate values allow our database to address a many-to-many relationship between a VIN and a car's location. Consequently, a new many-to-many relationship between a car's location and its utility companies emerged, requiring us to create a new 'car_location_id' and add an additional table to address this, differing from our initial design.

Ethics considerations

Personal Information Handling: The database contains personal information related to vehicle owners, including VINs, registration locations, and potentially DOL numbers. Ensuring the security and privacy of this sensitive information is paramount.

Compliance with Data Protection Regulations: Given the nature of the data, compliance with data protection regulations such as GDPR or similar regional laws is crucial. The project must follow legal frameworks to protect individuals' privacy rights.

Informed Consent: If the dataset includes personally identifiable information (PII), obtaining informed consent from vehicle owners for data collection, storage, and use is essential to uphold privacy standards.

Purpose Limitation: Ensure that the data collected is used solely for the intended purpose of analyzing trends and growth in the electric vehicle market. Avoid any use that may infringe upon the privacy or rights of individuals.

Transparency and Accountability: Clearly communicate the project's objectives, data usage, and methodologies to stakeholders. Establish accountability mechanisms to address any concerns related to fair use.

Data Accuracy: The database must prioritize data accuracy to provide reliable insights. Regular updates are necessary to ensure stakeholders have access to current and relevant information.

Benefit to Stakeholders: The project's focus on the increasing importance of electric vehicles and potential benefits for stakeholders is positive. However, ethical considerations should ensure that the benefits are distributed equitably and do not harm any particular group or community.

Security Measures: Implement robust security measures to safeguard the database from unauthorized access. This is critical to maintaining the confidentiality and integrity of the data.

Lessons Learned

Initially, we mistakenly believed that the dataset contained errors due to multiple occurrences of the same VIN number in different locations. Consequently, our original plan involved filtering the data to include each VIN only once. However, after discussions with the teaching assistants, we came to the realization that the dataset, in fact, did not contain errors. Instead, we needed to confront the inherent many-to-many relationship between cars and locations. Subsequently, it became apparent that cars and locations also shared a many-to-many

relationship with utility companies. Additionally, upon importing the data, it became clear that the creation of a 'car_location_id' was necessary. This was essential to address duplicate values for cars that are registered in the same location but with distinct department of licensing numbers. These experiences proved valuable lessons, enhancing our understanding of the dataset and guiding the development of a more robust database to effectively tackle these complexities.

Potential future work

Given the current time constraints and submission guidelines emphasizing minimal number of rows, the existing database may not fully capture the nuances of the electric vehicle dataset. With more time, an important addition would involve significantly expanding the dataset by adding many additional rows to enhance accuracy. Furthermore, in the future it would be wise to accommodate the continuous influx of new electric vehicle models by creating new 'model_id' entries and incorporating them into the 'models' table as new electric vehicles become available. As the electric vehicle landscape evolves, it would be beneficial to extend the 'battery' table to include new types of vehicles reflecting the diverse advancements in this area.

Citations

Washington State Department of Licensing. (2023). Electric Vehicle Population Data. URL: https://data.wa.gov/Transportation/Electric-Vehicle-Population-Data/f6w7-q2d2/about_data